

Following completion of the preparation materials, students should be able to:

1. Compare and contrast the within-class and between-class pharmacokinetic properties of the antiretroviral agents.
2. Describe the structure-activity relationships, mechanism of action, and mechanisms of viral resistance of each class of antiretroviral agents in relation to specifically targeted viral proteins and give examples.
3. Explain the clinical rationales for selection of specific antiretroviral therapy (ART) and give examples for:

ART-naïve patients	Transgender patients
Switch to 2-drug ART	Women (assigned female at birth)
Treatment-experienced patients	Pregnant people and peripartum care
Pre-exposure prophylaxis (PrEP)	Neonates
Post-exposure prophylaxis (PEP)	Pediatric patients
	Older patients
4. Differentiate the class and drug-specific adverse effects and their impact on patient adherence to therapy.
5. Explain the mechanism-related and metabolism- and transport-related drug interactions for specific drugs, including the antiretroviral drugs that should not be used in combination.

Characteristics	DOR	EFV	RPV*
Dosing	Once daily	Once daily	Once daily
Food requirement	With or without food	Empty stomach	With food (needs acid pH)
Single drug tab Fixed-dose combos	yes DOR-TDF-emtricitabine	yes EFV-TDF-emtricitabine	yes RPV-TDF-emtricitabine RPV-TAF-emtricitabine
Adverse effects	Generally well tolerated	Neuropsych side effects Dizziness, abnormal dreams, headache depression, insomnia, more Skin rash QTc prolongation	Generally well tolerated Depression Headache QTc prolongation
CYP3A4 DDIs	CYP3A4 substrate	CYP3A4 substrate Mixed inducer / inhibitor	CYP3A4 substrate
Other significant DDIs	None	Inducer of CYP2B6, 2C19	PPIs: ↑ gastric pH → ↓ absorption → avoid

*Rilpivirine: Treatment-naïve adults with HIV RNA $\leq 100,000$ copies/ mL and CD4 count ≥ 200 c/mm³ due to increased risk of virologic failure in patients with baseline viral load $> 100,000$ copies/mL.

NNRTIs: HIV Resistance | Adverse Effects | Drug Interactions

Resistance

- **Single amino acid changes** in binding pocket confers *high-level drug resistance*.
- **High-level cross-resistance** occurs within the NNRTI class.
- **Transmitted resistance** of NNRTI-resistant HIV is more common than with INSTIs and PIs.

Resistance testing should be performed before initiation of NNRTI-based ART, even in ART-naïve patients.

Rashes:

- Occur frequently with all NNRTIs, usually during the first 4 weeks of therapy
 - Usually mild and self-limited
 - SJS/TEN rarely
- GI disturbances

Drug interactions: High potential

- Substrates and inducers or inhibitors of metabolic enzymes
- Efavirenz-methadone → CYP2B6, 3A4 induction may ↓methadone levels and lead to opioid withdrawal symptoms

NNRTIs What you should know about nevirapine and etravirine

Nevirapine

- Low barrier to resistance
- Not recommended for initial ART
- Mainly used as an option for presumptive therapy in **neonates** at high-risk of HIV:
ZDV + emtricitabine + nevirapine
- AEs: Hepatotoxicity, hepatic necrosis, rash, SJS/TEN/organ dysfunction
- NVP should not be used in females with CD4 count >250 cells/mm³ or males >400 cells/mm³ due to association with severe, life-threatening, and fatal hepatotoxicity.

Etravirine

- Higher barrier to resistance
- Option for ***treatment-experienced patients*** (if the drug/HIV is susceptible)
- Reduced toxicity compared to efavirenz
- May cause rash
- CYP3A4 substrate and inducer
- CYP2C9, 2C19 inhibitor

Questions to Review the Key Points

1. What are the important (clinically relevant) take-aways about the PK properties of the NNRTIs class?
2. How is viral resistance related to NNRTIs' mechanism of action?
3. What criteria must be met before rilpivirine is administered to the ART-naïve patient?
4. What are the class and drug-specific toxicities to remember?
5. Which patients are at highest risk for nevirapine-related hepatotoxicity?

HIV Protease Inhibitors

- * Atazanavir, boosted
- * Darunavir, boosted

Activity: HIV-1

Darunavir has more activity against HIV-2 than some other PIs.

PI-based regimens have demonstrated virologic potency, durability in treatment-naïve patients, and a high barrier to resistance.

PIs have are associated with numerous side effects and drug interactions.

Lopinavir/r, fosamprenavir/r, saquinavir/r have lower efficacy, greater pill burden, or increased toxicity and are not recommended for initial treatment in ART-naïve patients.

Characteristics of PIs

Characteristics	ATVc or ATVr	DRVc or DRVr
Dosing with food	Once daily	Once daily: PI-naïve Twice daily: Treatment-experienced pt
Fixed-dose combos	ATVc	DRVc and DRVc-TAF-emtricitabine
Adverse effects: Class: nausea/vomiting/diarrhea, cardiac conduction abnormalities, glucose and lipid metabolism – Cushingoid fat redistribution, ↑LDL, ↑TG, hyperglycemia, insulin resistance		
Drug-specific adverse effects	Jaundice ↑ Indirect bilirubin (benign) Cholelithiasis Nephrolithiasis Kidney injury	Skin rash (can be severe) Caution in patients with h/o sulfonamide allergy. Increased serum transaminases Liver toxicity, including hepatitis, reported. Contraindicated in severe liver disease
CYP3A4 DDIs	CYP3A4 substrate, inhibitor	CYP3A4 substrate, inhibitor
Other significant DDIs	PPIs, other drugs that ↑ gastric pH → ↓ absorption → separate administration	Not applicable

Pregnancy: Boosted ATV or DRV twice daily are preferred PIs for initial therapy in all trimesters.¹⁶

PIs: High Potential for Drug Interactions

Problem	Relevant examples
CYP3A4 inhibition → Elevated levels of substrate drugs	Drugs with narrow therapeutic window • Statins: ↑ levels → ↑ muscle toxicity risk (rhabdomyolysis, acute kidney injury if severe) • Midazolam: narrow therapeutic window • Many, many more
P-glycoprotein inhibition	Digoxin (cardiotonic) is a substrate of P-glycoprotein; inhibition of P-gp → ↑ risk of digoxin toxicity
Proposed mech: UGT induction by ritonavir/PI	Hormonal (estrogen) contraceptive decreased efficacy; use alternative method of contraception ¹
Unknown mechanism	Methadone levels are decreased → may induce opioid withdrawal syndrome ²

Questions

1. What are the most commonly used PIs in ART for initial therapy?
2. What is the rationale for concomitant administration of ritonavir or cobicistat?
3. What adverse effects are associated with the PIs class?

Maraviroc pharmacology

PK	Oral, variable bioavailability 23-33%, CYP3A4, feces/urine $t_{1/2}$ 14-18 hours → 2x daily dosing
MOA	Blocks binding of the HIV outer envelope protein gp120 to the CCR5 chemokine receptor
Activity	Activity: CCR5-tropic strains of HIV only
Resistance	1)Shift in tropism to CXCR4 or dual tropism 2)Mutations in the V3 loop of gp120 that allow virus binding in the presence of inhibitor
Prior to therapy, tropism testing should be performed for presence of CCR5-tropic HIV-1 infection. Therapy not recommended in patients with CXCR4- or dual/mixed tropic HIV-1 infection; efficacy not demonstrated in this population.	
Uses	Treatment of CCR5-tropic HIV-1 infection ≥16 years old with other ARV drugs
Adverse effects	Hepatotoxicity: May be preceded by severe rash or evidence of systemic allergic reaction
Drug interactions	CYP3A4 strong inhibitors and inducers Consider therapy modification

Enfuvirtide Pharmacology

Chemistry	Synthetic peptide (36 amino acids) derived from a part of the transmembrane gp41 protein of HIV-1 involved in the fusion
PK	SubQ; Metabolized by proteolytic hydrolysis, $t_{1/2}$ ~4 hours → 2x daily
MOA	Binds HIV envelope protein gp41 Inhibits the conformational change in gp41 required for membrane fusion → Inhibits the fusion of HIV-1 virus with CD4 cells
Resistance	Mutations in the codon of the enfuvirtide binding domain of gp41
Use	Treatment-experienced: Adults and children ≥ 6 years old Evidence of HIV replication despite ongoing antiretroviral therapy Combination with at least two other active antiretroviral drugs
Adverse effects	Frequent local cutaneous reactions (injection site) pain, erythema, induration, and nodules and twice daily dosing make long-term use difficult.
Drug interactions	No known metabolic interactions

PK	Prodrug, oral → Hydrolysis to temsavir CYP3A4 metabolism, P-gp substrate → High potential for drug interactions. Strong CYP3A4 inducers should be avoided.
MOA	Binds HIV-1 gp120 envelope adjacent to the gp120-CD4 binding site → prevents the gp120 conformational change → prevents attachment of CD4
Use	Treatment of HIV-1 infection in heavily treatment-experienced adults with multidrug-resistant HIV-1 infection failing their current ART due to resistance, intolerance, or safety considerations.
Adverse effects	Nausea (most common) Elevations of hepatic enzymes, esp. in HBV, HCV coinfection QT interval prolongation
Drug interactions	Strong CYP3A4 inducers are contraindicated.

Lenacapavir PK, Uses, and Adverse Effects

A	Oral initiation subQ maintenance
D	Protein binding >98% Widely distributed – very large Vd
M	Complex, including CYP3A

E	Feces, unchanged drug and metabolites
t _{1/2}	Oral: 10-12 days subQ: 8-12 weeks
	Oral and subQ loading dose schedule Maintenance: subQ once every 6 months

Therapeutic use

- HIV-1 in heavily treatment-experienced adults with multidrug-resistant HIV-1 infection failing their current antiviral therapy due to resistance, intolerance, or safety concerns
- Use in combination therapy with other antivirals.

Adverse effects

- Generally well tolerated. Injection site reactions frequently reported
- **Contraindication:** Concomitant CYP3A4 inducers

Initial ART (Treatment-Naïve Patients)

2 NRTIs *plus* an INSTI

Alternatives: an NNRTI *or* a boosted PI

1. Tenofovir / Emtricitabine
 - fixed-dose combination with another ARV
 - 3-drugs in one pill, once daily
2. Abacavir with Lamivudine
 - For HLA-B*5701-negative patients

 **Resistance testing** should be used as a guide to selection of an antiviral regimen in both treatment-naïve and treatment-experienced individuals.

Treatment-experienced patients

- Conduct resistance testing while the patient is taking a failing regimen
- Goals of treatment:
 1. To reestablish maximal virologic suppression, with HIV-RNA suppressed below the limit of detection of a sensitive assay
 2. If maximum virologic suppression is not possible, then preserve CD4 cell count, avoid progression, minimize toxicities
- Select ART with activity against the drug-resistant viral strains
 - 3 active drugs → more sustained response
- Assess adherence frequently and simplify the regimen when feasible

Pre-Exposure Prophylaxis (PrEP)

For prevention of HIV-1 infection in adults who are at very high risk for acquiring HIV

- such as individuals who engage in sexual activity within a high prevalence area or social network or intravenous drug users

PrEP as part of a comprehensive prevention strategy

- confirm negative HIV-1 test prior to starting PrEP
- reconfirm for HIV-1 infection at least every 6 months while the patient receives PrEP
- comprehensive prevention strategy, including the importance of adherence, risk reduction counselling, ready access to condoms

- TDF-Emtricitabine orally daily for men or women (long experience with this drug)
- TAF-Emtricitabine orally daily for adolescents, transgender women, and men with bone and renal problems who are not eligible for TDF
- Cabotegravir I.M. in gluteal muscle every 2 months for adults (no renal or lipid monitoring is required)

Transgender People with HIV

- ART is recommended for all transgender people living with HIV to improve their health and reduce the risk of HIV transmission to HIV-uninfected sex partners.
- HIV care services should be provided within a gender-affirmative care model.
- Prior to ART initiation, a pregnancy test should be performed for individuals of child-bearing potential.
- Some ARV drugs may have pharmacokinetic interactions with gender-affirming hormone therapy. Clinical effects and hormone levels should be routinely monitored with appropriate titrations of estradiol, testosterone, and androgen blockers as needed.
- Gender-affirming hormone therapies are associated with hyperlipidemia, elevated cardiovascular risk, and osteopenia. Therefore, clinicians should choose ART regimens that will not increase the risk of these adverse effects.

Women with HIV (assigned female at birth) (non-pregnant and using hormonal contraception)

Effects on hormone levels:	Transmission, acquisition of HIV:	Intrauterine devices (IUDs):
PIs / NNRTIs are inducers and/or inhibitors of hormones: • Inducers ↓ hormone levels → risk of pregnancy • Inhibitors ↑ hormone levels → risk of toxicities	Regardless of hormonal contraceptive use: • Consistent use of condoms (male or female) during sex • Adhere to an HIV regimen that effectively maintains viral suppression.	Appear to be a safe and effective contraceptive options for HIV-infected women.

Pregnant People with HIV

Goal: To maintain a viral load below the limit of detection throughout the pregnancy to reduce transmission to the fetus and newborn.

All HIV-infected pregnant people should receive a combination antepartum antiretroviral (ARV) drug regimen, regardless of CD4 count or HIV RNA copy number.

Therapy selection considerations

- Extensive experience with the drugs in pregnancy
- Well tolerated
- Good short-term safety profile
- PK not significantly altered during pregnancy
- High placental transfer to fetus
- No evidence of teratogenicity

ART:

Preferred:

2-NRTI backbone plus Dolutegravir

Alternative 3rd drug:

INSTI: Bictegravir, Raltegravir

PI: Atazanavir, boosted

NNRTI: Efavirenz, Rilpivirine

Others are not recommended due to limited safety or dosing information.

Prevention of Intrapartum Mother-to-Child Transmission

Zidovudine I.V. during labor if patient has HIV RNA $>1,000$ copies/mL (or unknown HIV RNA) near delivery

Zidovudine is not required for HIV-infected women receiving combination ART who have HIV RNA $\leq 1,000$ copies/mL consistently during late pregnancy and near delivery and no concerns regarding adherence to the regimen.

Unknown HIV status

- Rapid HIV screening test
- If positive, confirmatory testing
- Initiate mother and infant HIV drugs pending results
- If HIV test results are negative, stop infant ARV therapy

Neonate Prophylaxis of infection

Management in the newborn is based on risk of HIV infection and depends on the mother's use of antepartum antiretroviral agents and viral load within four weeks of delivery.

Low risk: Zidovudine (ZDV) x 4 weeks

- Infants ≥ 37 weeks gestation; maternal ART ≥ 10 weeks and achieved or maintained viral suppression (< 50 copies/mL) for the remainder of the pregnancy and no concerns related to adherence
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Infants that do not meet the above criteria or the high-risk criteria:
ZDV for 4 to 6 weeks

High risk: Presumptive ART from birth for 2 to 6 weeks

ZDV + emtricitabine + nevirapine, or ZDV + emtricitabine + raltegravir

- A newborn whose mother did not receive antepartum drugs *or* received only intrapartum drugs *or* did not have viral suppression within 4 weeks of delivery *or* had acute or primary HIV during pregnancy.

Treatment of HIV infection in Pediatric Patients

Preferred: INSTI-based ART regimen

- Adherence can be difficult in children and the child's growth and development can affect HIV treatment

Not recommended:

- Efavirenz-based regimens, etravirine-based regimens, maraviroc-based regimens, cabotegravir, 2-drug regimens, and ARV regimens with low efficacy or with insufficient data.

Breastfeeding

- Mothers with acute or primary HIV infection during breastfeeding should immediately discontinue breastfeeding.
- There is no consensus on appropriate management of ARV prophylaxis for infants of individuals with sustained viral suppression who are breastfed.

- NNRTIs are allosteric inhibitors of RT, inhibiting its polymerase activity. They do not require phosphorylation for activity.
- NNRTI resistance occurs rapidly with monotherapy. Etravirine has a high barrier to resistance compared to the other NNRTIs and is used for treatment-experienced patients.
- NNRTIs are associated with skin rash and GI disturbances. Efavirenz, rilpivirine, and, to a lesser extent, doravirine are associated with neuropsychiatric adverse effects. Nevirapine is associated with hepatotoxicity. Risk factors are females with CD4 >250 cells/microL and males with CD4 >400 cells/microL
- NNRTIs are substrates and inducers/inhibitors of CYPs. There is a high potential for drug interactions.

- HIV protease inhibitors (PIs) reversibly bind the enzyme, which prevents the cleavage of the viral polyprotein into reverse transcriptase, protease, integrase, and structural proteins. This action prevents viral maturation and results in the production of noninfectious virus.
- Resistance occurs with stepwise accumulation of mutations of the protease gene, leading to high-level resistance. Suboptimal PI concentrations result in more rapid appearance of resistant strains.
- PIs are extensively metabolized. They are substrates of CYP3A4 and excreted in urine as metabolites (very little unchanged drug). They are used with a pharmacokinetic enhancer (boosted), ritonavir or cobicistat, which are potent inhibitors of CYP3A4. There is a high potential for drug interactions
- GI adverse effects are common – nausea, vomiting, diarrhea. They cause hyperglycemia/diabetes, hypertriglyceridemia, hypercholesterolemia, fat accumulation in the abdomen and base of the neck and breast enlargement.
- Atazanavir/r and darunavir/r are preferred in the current guidelines due to improved adverse effect profile, virologic efficacy, and ease of dosing.

Entry inhibitors have diverse actions and effects.

- Maraviroc blocks CCR5 coreceptor that facilitates HIV entry through the host cell membrane. Viral tropism must be determined prior to initiating maraviroc therapy. Virologic failure is usually due to presence of CXCR4 coreceptor. Maraviroc is metabolized by CYP3A4. Therapy adjustments may be required in the presence of strong CYP3A4 inhibitors or inducers. Maraviroc is generally well tolerated. A concern is severe hepatotoxicity that may be preceded by a fever or rash. Liver function monitoring is recommended.
- Enfuvirtide a 36-amino acid synthetic peptide with a sequence derived from a part of the transmembrane gp41 protein of HIV-1. It binds gp41, preventing conformational change. It is administered by subcutaneous injection twice daily for treatment-experienced patients. Most adverse effects are related to the injection including pain, erythema, induration, and nodules, which occur in almost all patients.

- Fostemsavir oral prodrug is rapidly converted to temsavir, which is an attachment inhibitor. It binds directly to HIV-1 envelope gp120 near the gp120-CD4 site and prevents the gp120 conformational change and viral entry. It is for heavily treatment-experienced adults. CYP3A4 metabolism, half-life ~11 hours (twice daily dose). Strong CYP3A4 inducers should be avoided. The drug can cause elevations of hepatic enzymes, especially in patients with HBV, HCV coinfection, and QT interval prolongation.
- Ibalizumab is a post-attachment inhibitor by binding CD4, which interferes with the post-attachment steps required for HIV entry into host cells. IV infusion every 2 weeks for treatment of heavily ARV-experienced HIV-1 infected adults as part of a regimen of other potentially active agents whenever possible. It may cause dizziness, diarrhea, nausea, rash.
- Lenacapavir is a capsid inhibitor that interferes with several steps in the viral replication process used in combination with other ARVs for heavily treatment-experienced patients. It is generally well tolerated. Injection site reactions are frequent. Concomitant CYP3A4 inducers are contraindicated.